

G. Lejeune Dirichlet's Works. II.

XXIII.

**Excerpts from the Monthly Reports of the  
Academy of Sciences in Berlin**

Monthly Reports of the Academy of Sciences in Berlin, 1837–1853.

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of the Academy of Sciences in Berlin.**

**On the representation of arbitrary functions  
by definite integrals, with special application  
to the function  $P_n$  which occurs in the attraction of spheroids.**

(Monthly Report 1837, p. 79.)<sup>1</sup>

**On some problems  
which require the determination of an unknown function  
under the integral sign.**

(Monthly Report 1843, p. 152.)<sup>2</sup>

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<sup>1</sup>Only the title; cf. vol. I of this edition of G. Lejeune Dirichlet's Works, p. 285 and following. F.

<sup>2</sup>Only the title. F.

**Remarks on Kummer's proof of Fermat's theorem,  
concerning the impossibility of  $x^\lambda - y^\lambda = z^\lambda$  for an infinite number  
of prime numbers  $\lambda$ .**

(Monthly Report 1847, pp. 139-141.)

As regards the second of the two assumptions on which Mr. Kummer's ingenious proof rests, its correctness for each particular value of  $\lambda$  can be tested with the help of the general theory of complex units, of which some indications were given in the March report of 1846 and which will be published in one of the next issues of Crelle's Journal<sup>3</sup>. According to the theory just mentioned, for every  $\lambda$  one can set up the general expression for all complex units composed from  $\lambda$ -th roots of unity, and the formation of this expression presents no difficulty other than that of a numerical calculation whose complication increases rapidly with  $\lambda$ . Once this expression is known, containing  $\frac{\lambda-3}{2}$  fundamental units raised to indeterminate powers, it can soon be decided without great difficulty whether the condition stated in assumption (B.) is fulfilled; that is, whether the expression can be congruent, modulo  $\lambda$ , to a real number only when all exponents are divisible by  $\lambda$ .

Assumption (A.) concerns a theory which has the greatest analogy with that of quadratic forms. Just as not every number  $m$  for which the congruence

$$\xi^2 \equiv D \pmod{m}$$

is possible is always contained in the form  $x^2 - Dy^2$ , but in general a limited number of essentially different quadratic forms exists by which all such numbers  $m$  can be represented, so analogous relations occur between higher congruences and corresponding higher forms. Consider, for example, the congruence

$$\frac{\xi^\lambda - 1}{\xi - 1} \equiv 0 \pmod{m},$$

with respect to which Euler already completely determined the moduli  $m$  for which it is possible. These numbers  $m$  are likewise not always of the form

$$\varphi(\alpha)\varphi(\alpha^2)\dots\varphi(\alpha^{\lambda-1}),$$

where

$$\varphi(\alpha) = x_0 + \alpha x_1 + \alpha^2 x_2 + \dots + \alpha^{\lambda-2} x_{\lambda-2},$$

and  $x_0, x_1, \dots, x_{\lambda-2}$  denote indeterminate integers; but there always exist finitely many forms which, like the preceding one, can be decomposed into linear factors and which, together with it, express all numbers  $m$  for which Euler's congruence is solvable. After the representability of all units by  $\frac{\lambda-3}{2}$  fundamental units had been recognized, which here is the analogue of the general solution of Fermat's equation

$$x^2 - Dy^2 = 1,$$

it was natural to try to pursue the analogy between quadratic forms and these higher forms further, and in particular to determine the number of the latter by similar means to those by which the same question had earlier been settled in the theory of quadratic forms. This investigation, to which Mr. Kummer refers at the beginning of his communication, was in fact brought to completion about

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<sup>3</sup>The paper appeared in vol. 40 of the Journal, pp. 130-138. F.

three years ago with the help of a new principle not needed in the determination of the number of forms of the second degree, and led to a result whose form seems remarkable and which is as simple as could ever be expected in a question covering forms of all degrees. The expression that I found for the number of forms of degree  $(\lambda - 1)$ , which, as the analogy with quadratic forms led one to expect, contains the above-mentioned  $\frac{\lambda-3}{2}$  fundamental units, gives, as soon as these are known, the means of testing assumption (A.) by a fairly simple numerical calculation. Whether it will be possible, however, to derive something general about the connection between the two assumptions from the way in which the fundamental units enter into the expression for the number of forms, and to test the conjecture expressed by Mr. Kummer at the end of his letter, namely that the first assumption always involves the second, I do not venture to decide for the present. Such a decision can only be the result of a careful discussion of the expression for the number of forms, undertaken from the point of view just described.

**Report on Ehrenzeller's solution of numerical equations  
by series.**

(Monthly Report, 18 February 1850, p. 36.)

Mr. Dirichlet delivered a report on the writing, assigned to the class, of Mr. Ehrenzeller of St. Gallen, dated the 21st of December, containing a solution of numerical equations by series. The procedure agrees with the extension of Newton's approximation method to the term of the second order in the expansion according to Taylor's theorem, and therefore cannot lay claim to novelty.

**On a new derivation of two arithmetical theorems  
from a common source.**

(Monthly Report 1853, p. 300.)<sup>4</sup>

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<sup>4</sup>Only the title. F.